

Name: **Sara Haider** Reg. No: **F2023-744** Section No:**C**

Explore Private address spaces in IPv4 for each address class [CLO1]

Prepare a 2-page handout on:

Why were private IP address ranges introduced in IPv4 networks? [3]

How is security managed? [2]

Role of routers here? [2]

Examples [2]

Pros & Cons [2]

Check your home WiFi network. Find out your device IP address, is it private or public, find the default router (gateway) IP? [3]

[Clearly include sources/references in your submission](#) [1]

Solution:

1. Why were private IP address ranges introduced in IPv4 networks?

Solution:

Private IP address ranges were introduced to solve the problem of IPv4 address exhaustion. The IPv4 system provides about 4.3 billion unique addresses, which is insufficient for the rapidly growing number of internet devices.

To address this issue, special private address ranges were defined in RFC 1918. These addresses are used within local networks such as homes, offices, and organizations. Devices with private IP addresses cannot be accessed directly from the public internet.

Instead, they communicate with external networks through a router using Network Address Translation, which converts private IP addresses into a public IP address.

This approach allows many devices within a private network to share a single public IP address, helping conserve the limited IPv4 address space.

2. How is security managed?

Solution:

Security in private IP networks is managed mainly through network isolation and address translation. Devices using private IP addresses are not directly reachable from the public internet, which provides a basic level of protection. Routers use NAT to hide internal IP addresses when communicating with external networks. In addition, routers and firewalls can filter incoming and outgoing traffic, preventing unauthorized access to internal devices.

3. Role of routers here?

Solution: Routers play a critical role in private IP networks.

First, routers act as the default gateway that connects the local network to the internet. **Second**, routers perform Network Address Translation (NAT), which converts private IP addresses into a public IP address so that devices can communicate with external servers. Routers also manage network traffic and may assign IP addresses to devices using DHCP.

4.Examples

- **Solution:** 10.0.0.0 – 10.255.255.255 (Class A private range)
- 172.16.0.0 – 172.31.255.255 (Class B private range)
- 192.168.0.0 – 192.168.255.255 (Class C private range)

5.Pros & Cons

Solution: Advantages

- Conserves the limited IPv4 address space
- Provides basic network security by hiding internal devices
- Allows multiple devices to share one public IP address

Disadvantages

- Requires NAT configuration for internet access
- Some applications may experience connectivity issues due to NAT
- Troubleshooting network problems can sometimes become more complex

6. Check your home WiFi network. Find your device IP address, determine whether it is private or public, and find the default router (gateway) IP.

Solution:

After running the `ipconfig` command in Command Prompt, the following information was obtained from the Wi-Fi adapter.

Device IP Address:

192.168.1.6

Subnet Mask:

255.255.255.0

Default Gateway (Router IP):

192.168.1.1

The IP address **192.168.1.6** belongs to the **Class C private address range (192.168.0.0 – 192.168.255.255)** defined in **RFC 1918**.

Private IP address ranges are:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

```
C:\Windows\system32\cmd.exe
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::b837:3c1c:90e0:c83c%45
IPv4 Address. . . . . : 172.22.64.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::51bd:6074:481c:a3ef%4
IPv4 Address. . . . . : 192.168.1.6
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\3 Stars Laptop>
```

Since **192.168.1.6** falls within the **192.168.x.x** range, it is a **Private IPv4 Address**. The **Default Gateway 192.168.1.1** represents the router, which connects the local network to the internet and performs **Network Address Translation** to allow private devices to communicate with the public internet.

References:

<https://www.geeksforgeeks.org/computer-networks/private-ip-addresses-in-networking/>

<https://datatracker.ietf.org/doc/html/rfc1918>

<https://www.cisco.com/c/en/us/support/docs/ip/network-address-translation-nat/4606-8.html>

<https://www.techtarget.com/searchnetworking/tip/Introduction-to-IP-addressing-and-subnetting>